YEAR 8 MATHS · CHAPTER 5 · MIND MAP

# Algebra and index laws

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## 1 The language of algebra

In $3x + 5$: $x$ is a pronumeral, 3 is the coefficient, 5 is the constant.

"4 less than twice $n$" is $2n - 4$

## 2 Substitution and equivalence

Replace each letter with its value. Equivalent expressions are equal for every value.

$x = 3,\ y = -2$: $2x + y = 4 \quad xy = -6$

$2(x + 1)$ is equivalent to $2x + 2$

## 3 Adding and subtracting terms

Only like terms (same letters, same powers) can be added or subtracted.

$5a + 2b - a = 4a + 2b$

$3x^2 + x - x^2 = 2x^2 + x$

## 4 Multiplying and dividing terms

Multiply the numbers, then the letters. Divide like a fraction and cancel.

$3a \times 4b = 12ab \quad 2x \times 5x = 10x^2$

$12ab \div 3a = 4b$

**Watch out:** $x + x = 2x$ but $x \times x = x^2$

## 5 Algebraic fractions: + and −

Use a common denominator, then combine the numerators.

$\frac{x}{2} + \frac{x}{3} = \frac{3x}{6} + \frac{2x}{6} = \frac{5x}{6}$

$\frac{3a}{4} - \frac{a}{2} = \frac{3a}{4} - \frac{2a}{4} = \frac{a}{4}$

## 6 Algebraic fractions: × and ÷

Cancel common factors, then multiply. To divide, flip the second fraction.

$\frac{2a}{5} \times \frac{10}{a} = 4$

$\frac{a}{4} \div \frac{a}{8} = \frac{a}{4} \times \frac{8}{a} = 2$

## 7 Expanding brackets

Multiply every term inside by the term outside.

$3(x + 4) = 3x + 12$

$-2(x - 5) = -2x + 10$

$x(x + 3) = x^2 + 3x$

## 8 Factorising

The opposite of expanding: take out the highest common factor.

$6x + 9 = 3(2x + 3)$

$x^2 + 5x = x(x + 5)$

$12ab - 8a = 4a(3b - 2)$

## 9 Applying algebra

Use a letter for the unknown and write a rule.

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Rectangle, width $x$, length $x + 3$: perimeter $4x + 6$

## 10 Index laws: × and ÷

Same base: multiply, add the indices. Divide, subtract them.

$x^3 \times x^4 = x^7 \quad a^6 \div a^2 = a^4$

$2^3 \times 2^2 = 2^5 = 32$

**Watch out:** $2^3 \times 2^2$ is not $4^5$.

## 11 Zero index and power of a power

Anything to the power 0 is 1. Power of a power: multiply the indices.

$5^0 = 1 \quad 3x^0 = 3 \quad (x^2)^3 = x^6$